

AZURE ENTERPRISE INFRASTRUCTURE LAB

07 - Azure Alerts

Monitoring and Operational Alerting

Evidence-based implementation documentation with Azure Portal screenshots

1. Objective

Create and document an operational Azure Monitor alert for a meaningful VM condition. The project requirement calls for an alert rule, condition, action group/notification, and alert history.

Current status: The configuration and action-group creation are evidenced, but the supplied screenshots do not yet prove the final alert rule was saved or that an alert fired. Those validation items remain pending.

2. Resource

Setting	Value
VM	vm-web-MM
Monitoring	Azure Monitor
Alert type	Metric/recommended alert
Primary condition	Percentage CPU > 80%
Notification	Email
Estimated monthly alert cost shown	USD \$0.00

3. Implementation Flow

- vm-web-MM -> Azure Monitor Alerts
- Condition -> Percentage CPU > 80%
- Notification -> Email / Action Group
- Alert rule -> save and enable
- Alert history -> pending validation

4. Navigation Troubleshooting

The first screenshot opened a Microsoft Entra / Microsoft 365 group named Monitoring Alerts, rather than the Azure Monitor alert blade for the VM. The correct resource path is: Virtual machines -> vm-web-MM -> Monitoring -> Alerts.

5. Alert Configuration

Setting	Final intended value
Resource	vm-web-MM
Signal	Percentage CPU
Operator	Greater than
Threshold	80%

Notification	Email
Other recommended rules	Disabled
Estimated monthly cost	USD \$0.00

The intermediate configuration briefly had both Percentage CPU and Data Disk IOPS enabled. The corrected configuration screenshot shows only the CPU alert enabled and Data Disk IOPS disabled.

6. Action Group

The recommended-alert flow created an action group successfully. The Azure Portal notification explicitly confirms: Action group created successfully. The action group is intended to deliver email notification when the alert triggers.

7. Recommended Alert Rule Details

- Alert name: vm-web-MM-High-CPU
- Description: Alerts when the average VM CPU exceeds 80%.
- Enabled upon creation: Yes

8. Validation Plan

1. Confirm the CPU alert rule is saved and visible under vm-web-MM -> Monitoring -> Alerts.
2. Confirm the rule is enabled and the email/action group is attached.
3. Generate or wait for a CPU threshold breach during testing.
4. Capture the alert state and alert history.
5. Record the final fired/resolved status in this document.

9. Evidence Checklist

- ☒ Correct VM Alerts page
- ☒ CPU threshold condition configured
- ☒ Email notification enabled
- ☒ Action group creation success captured
- ☐ Alert rule saved successfully
- ☐ Alert fired during validation
- ☐ Alert history captured
- ☐ Final resolved/closed state captured

10. GitHub Structure

- 07-monitoring/README.md
- 07-monitoring/screenshots/
- 07-monitoring/screenshots/01-wrong-monitoring-alerts-page.png

- 07-monitoring/screenshots/02-vm-alerts-page.png
- 07-monitoring/screenshots/03-recommended-alerts-initial.png
- 07-monitoring/screenshots/04-cpu-alert-config-corrected.png
- 07-monitoring/screenshots/05-action-group-created.png

11. Git Commit

```
cd /d C:\azure-enterprise-infrastructure-lab
git add 07-monitoring
git commit -m "Document Azure Monitor CPU alert"
git push
```

12. Interview Skills Demonstrated

- Azure Monitor and metric alerts
- VM operational monitoring
- CPU threshold alerting
- Notification routing
- Action groups
- Alert lifecycle and history
- Azure Portal troubleshooting
- Cost-aware monitoring

13. Interview Questions

What is an Azure Monitor metric alert?

A rule that evaluates resource metrics against a threshold and can trigger an action when the condition is met.

Why use CPU > 80%?

It is a simple, meaningful operational signal for demonstrating threshold-based alerting on a VM.

What is an action group?

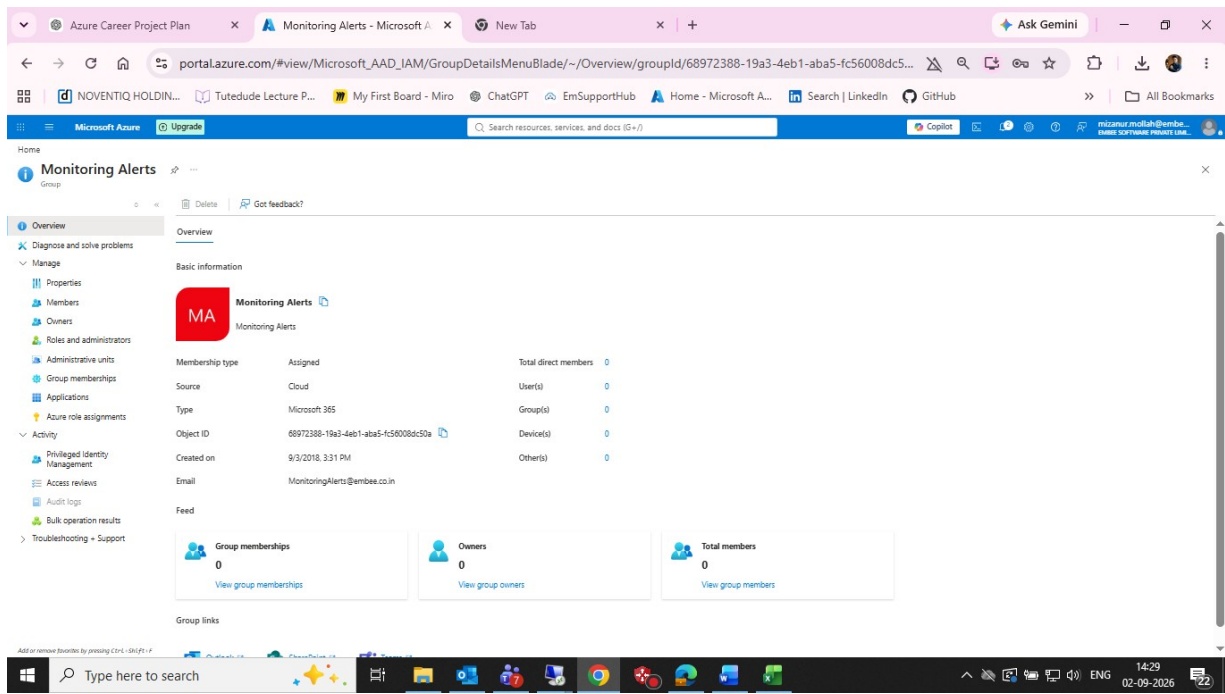
A reusable collection of notification/action settings used when an Azure Monitor alert is triggered.

Why document alert history?

It proves the alert was not only configured but also evaluated and triggered/resolved during validation.

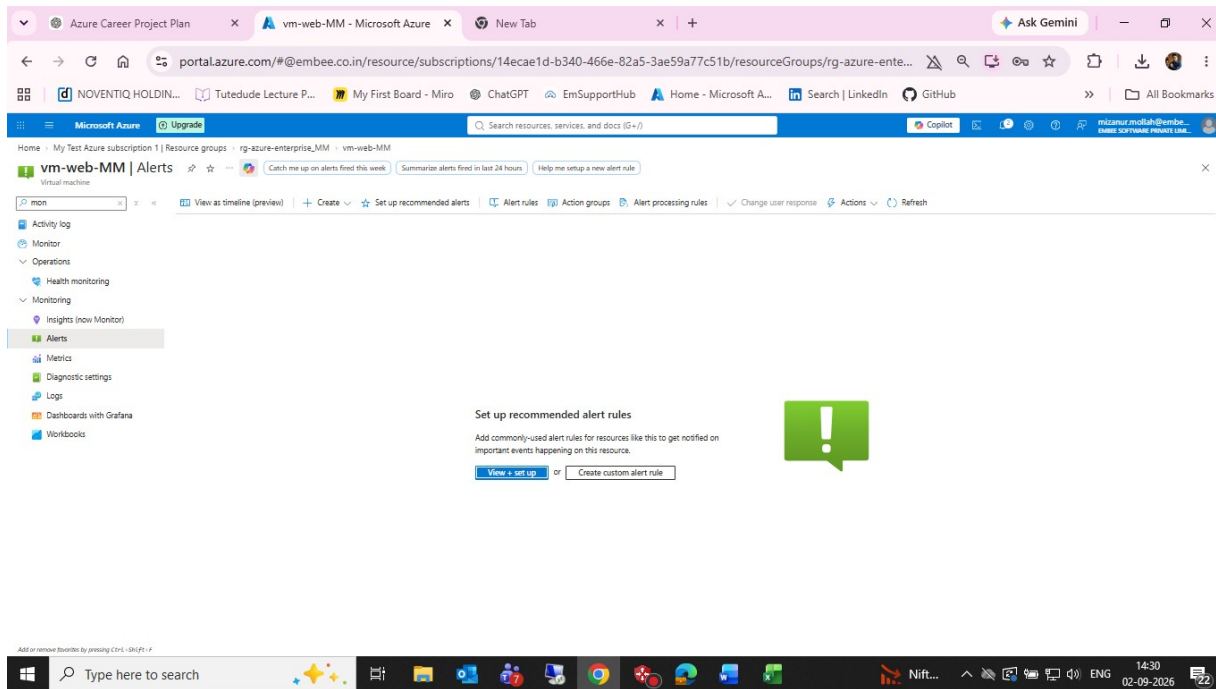
14. Evidence Screenshots

Figure 1 - Initial navigation - incorrect Monitoring Alerts group page (troubleshooting evidence)



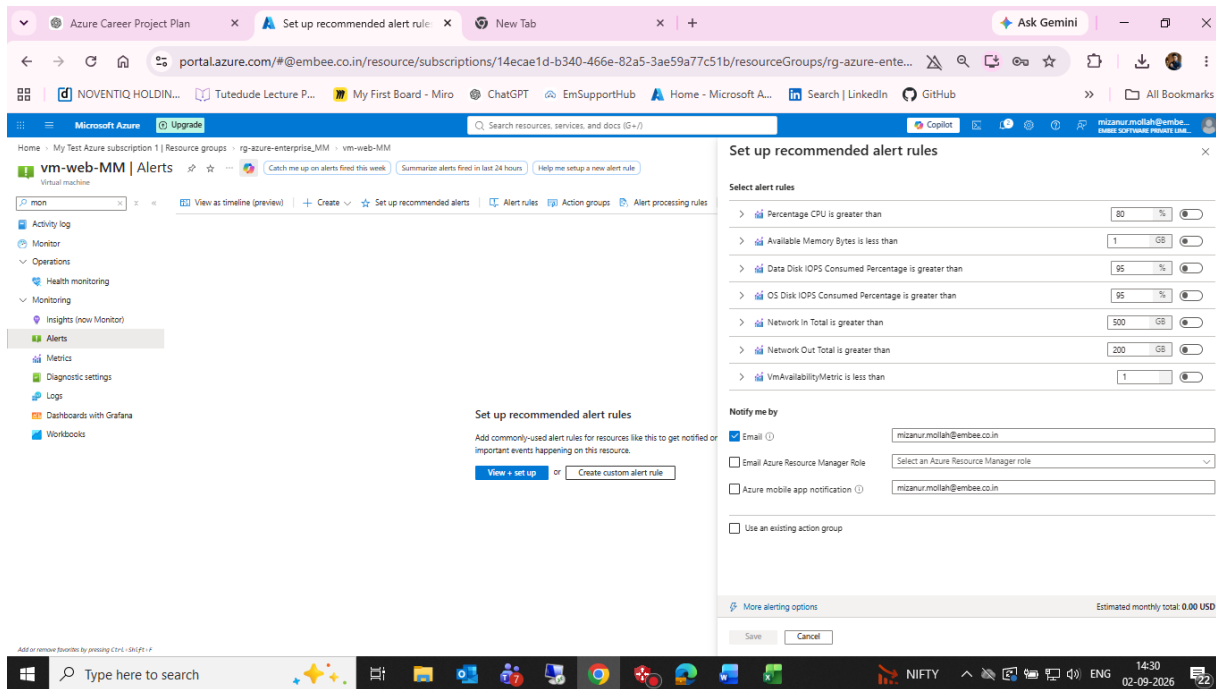
Initial navigation - incorrect Monitoring Alerts group page (troubleshooting evidence)

Figure 2 - Correct Azure Monitor Alerts page for vm-web-MM



Correct Azure Monitor Alerts page for vm-web-MM

Figure 3 - Recommended alert setup - initial configuration



Recommended alert setup - initial configuration

Figure 4 - Corrected alert selection - CPU alert on, Data Disk IOPS off, email enabled

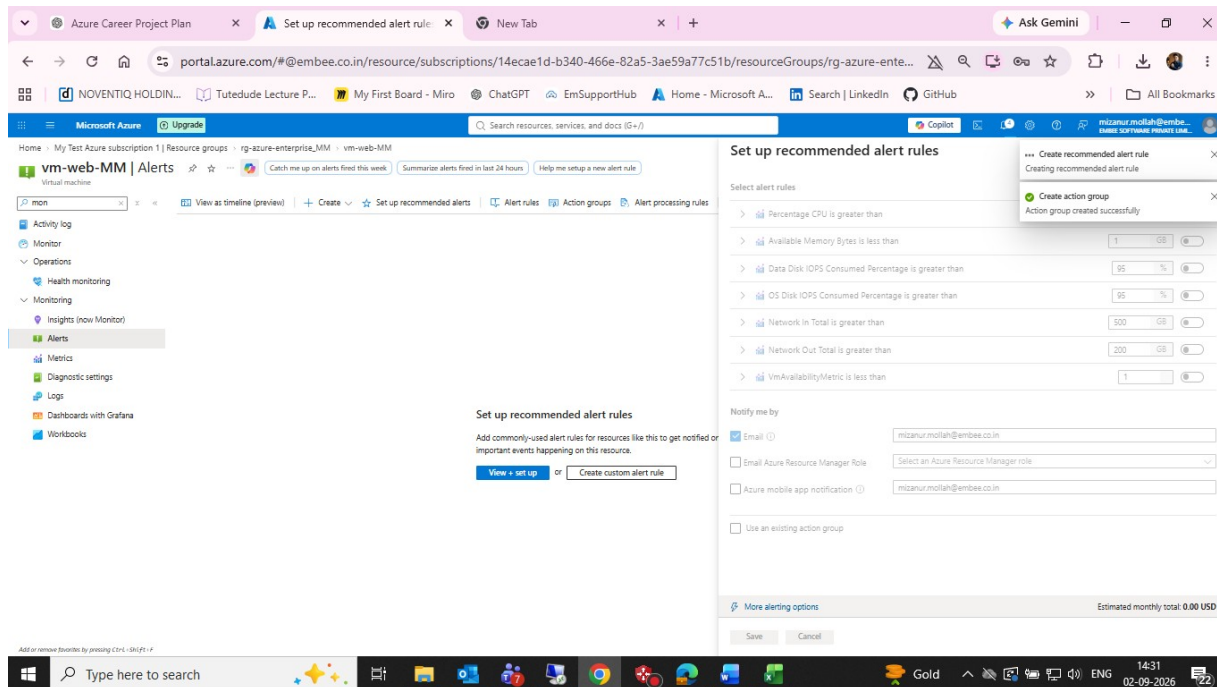
The screenshot shows the Microsoft Azure portal interface. On the left, the navigation pane is open, showing the 'Alerts' section under 'Monitoring'. The main area displays the 'Set up recommended alert rules' dialog for a virtual machine named 'vm-web-MM'. The dialog lists several metrics with their thresholds and status (on/off):

- Percentage CPU is greater than: 80 % (on)
- Available Memory Bytes is less than: 1 GB (off)
- Data Disk IOPS Consumed Percentage is greater than: 95 % (on)
- OS Disk IOPS Consumed Percentage is greater than: 95 % (off)
- Network In Total is greater than: 500 GB (off)
- Network Out Total is greater than: 200 GB (off)
- VmAvailability/Metric is less than: 1 (off)

Under the 'Notify me by' section, the 'Email' option is checked, and the email address 'mizanur.mollah@embee.co.in' is entered. The 'Email Azure Resource Manager Role' and 'Azure mobile app notification' options are unchecked. The 'Use an existing action group' option is also unchecked. At the bottom, there is a 'More alerting options' link and an 'Estimated monthly total: 0.00 USD'.

Corrected alert selection - CPU alert on, Data Disk IOPS off, email enabled

Figure 5 - Action group creation success notification



Action group creation success notification

15. Final Status

Task 07 is partially implemented. The correct VM alert blade is reached, the CPU threshold configuration is prepared, email notification is enabled, and the action group creation is confirmed. Final alert-rule creation and fired/resolved alert-history evidence are still required to mark the task fully complete.